



DPURemoteControl

Control the simulation state.

1. Inputs:

Start

If this input changes from 0 to 1 the following happens:
If the simulation is stopped the launch step is performed and the simulation is started afterwards.

Pause

If the simulation is in pause state it is resumed.

If this input changes from 0 to 1 and if the simulation is currently running it will be switched to pause mode.

Stop

If this input changes from 0 to 1 and if the simulation is currently running or if it is in pause mode it will be stopped.

Restart

If this input changes from 0 to 1 and if the simulation is currently running or if it is in pause mode it will be stopped and started again after the time given in *RestartDelay*.

Shutdown

If this input changes from 0 to 1 the simulation will be terminated.

2. Dynamic parameters:

RestartDelay

Time between Stop and Start of the simulation in millisec-onds.
Default: 3000

3. Static parameters:

OperatorIP

IP address of the operator computer

4. Functional principle:

DPURemoteControl establishes a network connection to the operator computer of a simulation and sends control commands. These control commands correspond to clicks on buttons of the toolbar in the SILAB GUI.

Control commands (like *Start*) can also be transmitted to the operator if the simulation is not currently running. Since under these conditions no DPU inputs and out-puts are synchronized via the network, the DPU with the output connected to the input *Start* of **DPURemoteControl** (the “controlling DPU”) has to be executed by the same computer and within the same thread by which/within which **DPURemoteControl** is executed.

Please note that if you develop the controlling DPU by yourself that it has to implement the method `cSPU::OnIdle(SPU_PREPARED)`. You can find more information on the topic in the documentation to the SILAB programming APIs.



5. Example:

The following script is a minimal example with which DPURemoteControl can be used:

```
DPUKeyboardState keyboard
{
    Computer={LOCALHOST};
    Index=1;
};

DPURemoteControlremoteControl
{
    Computer= {LOCALHOST};
    Index=1;
    OperatorIP="DESIRED_IP";
};

Connections=
{
    keyboard.S->remoteControl.Start,
    keyboard.X->remoteControl.Shutdown
};
```

When „S“ is pressed on the keyboard, the simulation on the machine with the IP-Address *DESIRED_IP* is started. When „X“ is pressed, the simulation is shut down.

These commands correspond to clicks on buttons of the toolbar in the SILAB GUI.